MEDORA

Integrated Clinic Management and Electronic Medical Records

System Design – Group 08

Hanoi University of Science and Technology

2026

Report review and system concept

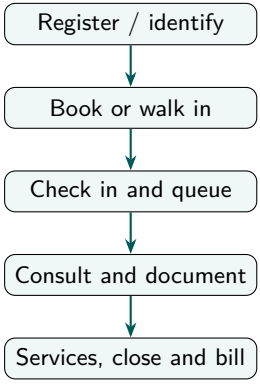
The outpatient workflow problem

Need

Clinic work spans registration, scheduling, queues, clinical notes, services and payment. Fragmented records and handoffs make the patient journey harder to coordinate and audit.

MEDORA's focus

One shared patient identity and encounter record, with explicit workflow states and role-based access.



System boundary and users

In scope

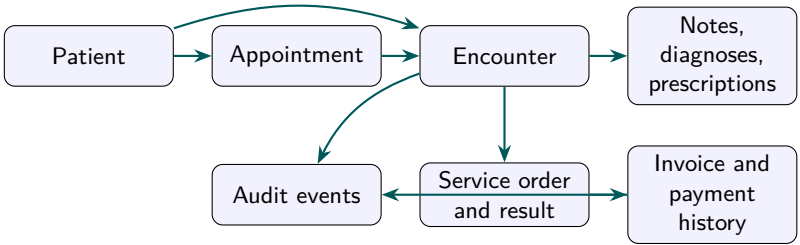
- Patient identity and longitudinal history
- Appointments, walk-ins, check-in and queues
- Screening, consultation, notes and prescriptions
- Service orders, execution, invoices and payments
- Reporting, audit trail and role/scope controls

Role boundaries

- Reception: identity, appointments, queues
- Nurse / doctor: screening and clinical decisions
- Technician / medicine staff: record service execution
- Cashier / finance: payments / authorized refunds
- Manager, admin and auditor: oversight and governance

Patient self-service, insurer settlement, external diagnostic systems and advanced inventory are later-stage extensions.

Core design: keep events and records distinct



Integrity rules emphasized in the report

A prescription is not a dispensing event; an order is not a result; signed notes are amended with a reason; refunds append a separate, authorized transaction.

Requirements and proposed acceptance targets

- Conflict-free booking under concurrent requests; no duplicate active encounter.
- Server-side authorization by role, facility scope and workflow state.
- Preserve signed clinical history; require accountable amendments and sensitive-action audit events.
- Idempotent payment handling and separately authorized refunds.

Targets stated by the report (not measured results)

25% lower median visit time; zero double-bookings in concurrency cases; at least 95% completion of mandatory encounter fields; complete audit coverage for tested sensitive events; rejection of all negative authorization cases.

Targets require a baseline, agreed workload and execution evidence before they can be evaluated.

Evidence status: design baseline, not a working product

Delivered in the report

Business and requirements baseline, traceable use cases, domain concepts, behavioral models and a 13-case test plan.

Not yet evidenced

No application, schema, API, deployment, test environment or user feedback is reported. All 13 test cases are Not Run; no pass/fail or performance result is claimed.

Open decisions: clinical warning sources, consent and retention, approval thresholds, branch scope and patient-portal policy.

Recommended next steps

1. Confirm clinic policies for clinical safety, authorization, corrections, finance and retention.
2. Approve architecture, persistence model and deployment assumptions.
3. Build a testable MVP for identity, booking/check-in, encounter notes, access control and audit evidence.
4. Run the planned tests with synthetic data; then extend prescription, service and finance workflows.

Questions?